Invoice & Document Processor

Product #2 — AI-Powered Extraction, Routing & QuickBooks Integration

PRODUCT	BUILD WINDOW	DATE
#2 of 4 — Auxo Suite	Weeks 3–4	March 2026

THE PROBLEM

Why Existing Tools **Fail SMBs**

Rossum, Nanonets, Docsumo — they all break in the same 3 ways. We fix all three.

5–15 hrs

per week — manual exception handling with current tools

$10–50K

implementation fees before a single invoice is processed

Template

based systems — break when a vendor changes their format

$0

our per-document fees — bundled into Auxo tier pricing

OUR EDGE

Three Fixes. **Zero Templates.**

LLM extraction handles any vendor format. Confidence routing eliminates manual review. Tier pricing kills per-doc fees.

Automated Triage

Confidence-based routing auto-posts high-certainty invoices and surfaces only real exceptions. Cuts manual review from 5–15 hrs/week to near zero.

No Templates Needed

Claude extracts from any vendor format — no template setup, no breakage when vendors change invoices. Works on day one with every supplier.

SMB-Friendly Pricing

Bundled into existing Auxo tiers ($1,500–$5,000/mo). No per-document fees, no $10K implementation just for invoicing.

Native QuickBooks Sync

OAuth once, sync forever. Auto-posts validated invoices as QB Bills. Batch support for high-volume clients.

INFRASTRUCTURE

Tech **Stack**

Open-source where it matters. Best-in-class where it counts. All research-validated.

LAYER	TOOL	WHY
PDF Parsing	**Docling (IBM)**	97.9% table accuracy, Apache 2.0, handles PDF/DOCX/images
OCR Fallback	**Surya OCR**	97.7% accuracy on invoices, 90+ languages, open-source
Extraction	**Claude 3.5 Sonnet**	Best JSON consistency, ~$0.001–0.01/invoice, any vendor format
Validation	**Pydantic + Thresholds**	Type-safe schemas, confidence routing at 0.80/0.95 cutoffs
Accounting	**QuickBooks Online API**	OAuth 2.0, batch endpoints — where most SMBs already live
Backend	**Python (FastAPI)**	Matches Derra AI stack, fast async processing
Frontend	**React + Vite + Tailwind**	Standalone client portal — review queue + dashboard
Deployment	**Railway**	Consistent with all Auxo/Derra deployments

Cost Per Invoice

Docling + Surya = free (open-source). Claude extraction = $0.001–0.01 per invoice. At 500 invoices/month per client, total AI cost is $0.50–$5.00/month. Margin on a $1,500/mo client: **>99%** on this product alone.

ARCHITECTURE

Processing **Pipeline**

Document in, validated bill in QuickBooks out. Seven steps, fully automated.

1. **Document Upload**
   Client drags & drops invoices in the portal. Supports PDF, DOCX, PNG, JPG. Email ingestion TBU.

2. **Document Parsing (Docling)**
   IBM's Docling extracts text, tables, and layout structure. Handles multi-page invoices with built-in document splitter and table merging.

3. **OCR Fallback (Surya)**
   If document is scanned or image-heavy, Surya OCR processes it at 97.7% accuracy. Supports 90+ languages and European number formats.

4. **Structured Extraction (Claude)**
   Claude 3.5 Sonnet extracts all invoice fields into a typed Pydantic schema. No templates — handles any vendor format. Returns confidence scores per field.

5. **Confidence Router**
   Scores each extraction. >0.95 = auto-post to QB. 0.80–0.95 = review queue. <0.80 = flagged for manual entry with pre-filled data.

6. **QuickBooks Integration**
   Maps validated invoice to QB Bill format. Posts via OAuth 2.0 with batch support (up to 25 at once). Proactive token refresh.

7. **Client Portal**
   Review queue for flagged invoices, audit log of all processed documents, dashboard with metrics and time-saved estimates.

CORE DIFFERENTIATOR

Confidence-Based **Routing**

The key innovation. Every invoice gets a confidence score. Routing is automatic.

> 0.95

Auto-Post

Directly posted to QuickBooks as a Bill. No human intervention. Target: 80%+ of all invoices land here.

0.80–0.95

Review Queue

Appears in client portal with uncertain fields highlighted. One-click approve or edit & approve.

< 0.80

Manual Entry

Pre-filled form with whatever was extracted. Client completes missing fields. Still faster than starting from scratch.

EXTRACTION SCHEMA

What We **Extract**

Typed Pydantic schema. Every field has a confidence score.

```
// Core invoice fields extracted by Claude

vendor_name: string
vendor_address: string
invoice_number: string
invoice_date: date
due_date: date | null
line_items: LineItem[] // description, qty, unit_price, amount
subtotal: decimal
tax: decimal
total: decimal
currency: string // USD, EUR, GBP, etc.
payment_terms: string | null // Net 30, Due on Receipt, etc.
confidence_score: float // 0.0 – 1.0 overall
```

MVP FEATURES

Five Core **Capabilities**

Everything needed to process invoices end-to-end. Nothing more.

FEATURE 1

Document Upload

Drag & drop in the client portal. PDF, DOCX, PNG, JPG. Batch upload supported. Email ingestion to be added post-MVP.

FEATURE 2

AI Extraction Engine

Docling parses structure. Surya OCR handles scanned docs. Claude extracts to typed schema. No templates, no vendor setup — works with any invoice format from day one.

FEATURE 3

Confidence Routing

Auto-post (>0.95), review queue (0.80–0.95), or flag (<0.80). The differentiator vs every competitor — eliminates 5–15 hrs/week of manual triage.

FEATURE 4

QuickBooks Integration

OAuth 2.0 setup once per client. Maps invoices to QB Bills. Batch posting (up to 25). Proactive token refresh + sync confirmation log.

FEATURE 5

Client Portal (Standalone)

Review queue with highlighted uncertain fields. Audit log of every invoice processed. Dashboard: volume, auto-post rate, time saved. Standalone auth — independent from Derra AI.

COMPETITOR ANALYSIS

Pitfalls We **Avoid**

Documented failure modes from Rossum, Nanonets, and Docsumo. We designed around every one.

PITFALL	OUR FIX
Template overfitting	Claude LLM handles any vendor format — no templates to create or maintain
Multi-page invoice failure	Docling has built-in document splitter + multi-page table merging
Foreign currency / number formats	Claude natively understands European decimal/thousands separators
OAuth token expiry	Proactive token refresh + client alert on auth failure before it blocks processing
Silent accuracy drift	Confidence score logged per vendor — alerts if any vendor's accuracy drops over time
SMB pricing barrier	Bundled into Auxo tier pricing — no per-document fees, no $10K implementation

BUILD SEQUENCE

Two-Week **Sprint**

Ship in 10 working days. No scope creep.

Day 1–2

Foundation

Repo setup, FastAPI skeleton, Docling + Surya integration, Pydantic invoice schema, file upload endpoint.

Deliverable: Parse any PDF to structured text

Day 3–4

Extraction + Routing

Claude extraction logic, confidence scoring, routing engine. End-to-end: upload PDF, get structured JSON + route decision.

Deliverable: Full extraction pipeline working

Day 5

QuickBooks OAuth + Sync

QB OAuth 2.0 flow, bill creation endpoint, batch posting, sync confirmation log.

Deliverable: Auto-post invoice to QB sandbox

Day 6–8

Client Portal

React frontend: review queue, audit log, dashboard, settings. Standalone authentication.

Deliverable: Functional client portal

Day 9–10

Test + Deploy

End-to-end testing with 20 real invoices. QB sandbox sync verification. Deploy to Railway.

Deliverable: Production-ready, deployed

VERIFICATION

How We Know **It Works**

Concrete targets. Tested with real invoices before any client touches it.

>80%
Auto-post rate — invoices that go straight to QB with zero human touch

<5 sec
Average processing time per invoice — upload to extraction complete

20
Real invoices tested — mix of PDF, scanned, multi-page, foreign currency

100%
QB sync verification — every auto-posted invoice confirmed in sandbox

KEY DECISIONS

Locked **In**

These are resolved. No revisiting.

ACCOUNTING PLATFORM
QuickBooks First
Most SMB clients already use QB. Xero added later.

EMAIL INGESTION
TBU
Evaluating approach. Skipped for MVP — portal upload covers launch.

PORTAL AUTHENTICATION
Standalone
Independent from Derra AI. Own auth system for client portal.

Revenue Impact
Invoice processing adds $500–800/mo per client as an add-on. For accounting and medical billing clients, this is the highest-value tool in the Auxo suite.

Integration with Product Suite
Bundles into Growth ($2,500/mo) and Scale ($5,000/mo) tiers. Standalone sale also possible for accounting-only clients via RCN channel.

Ship It. Week 3.

Upload an invoice. Get a QB Bill. No templates. No manual entry.

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